

1 a an invalid ISBN is one that is not a multiple of 11 ($\not\equiv 0 \pmod{11}$)
let our modified ISBN have modified digit j , with value m_j

as the digit sum is unaffected by this modification for all $i \neq j$

the difference between the old ISBN & the new is $j(d_j - m_j) = \delta$

in order for this new ISBN to be valid δ must be a multiple of 11 assumed towards contradiction

as $\text{new} = \text{old} - \delta$ and $\text{old} \equiv 0 \pmod{11}$, so for $\text{new} \equiv 0 \pmod{11}$, their difference must divide 11 $\delta \equiv 0 \pmod{11}$

$\delta = j(d_j - m_j)$ $0 \leq j \leq 9$ as 11 is prime either j or $(d_j - m_j)$ must be a multiple of 11
 $0 \leq d_j, m_j \leq X$

j cannot, as its maximum value is 9, likewise $d_j - m_j$ has a maximum of X when $d_j = X$ & $m_j = 0$
& a minimum of $-X$ when $d_j = 0$ & $m_j = X$

this range contains no multiples of 11, so $(d_j - m_j) \not\equiv 0 \pmod{11} \forall d_j, m_j \in \{0, \dots, X\}$

this contradicts our assumption so $\delta \not\equiv 0 \pmod{11}$, thus our new ISBN $\not\equiv 0 \pmod{11}$ & is thus invalid

b let $a, b \in \{0, \dots, X\}$ & $i \in \{0, \dots, 9\}$

i is the index of the digit with value a , & $i+1$ for b , these are digits of I

Validity conditions

$$\sum_{\substack{j=0 \\ j \neq i, i+1}}^{10} j \cdot d_j + i \cdot a + (i+1)b \equiv 0 \pmod{11} \text{ given that } I \text{ is valid}$$

Swapped

$$\sum_{\substack{j=0 \\ j \neq i, i+1}}^{10} j \cdot d_j + i \cdot b + (i+1)a \equiv 0 \pmod{11} \text{ assume towards contradiction}$$

set equal, and cancel all other digits

$$i \cdot a + b \cdot i + b \equiv i \cdot b + i \cdot a + a \pmod{11}$$

$b \equiv a \pmod{11}$ this contradicts the given $a \neq b$, thus the ISBN post swap cannot be valid

2 FLT $a^{p-1} \equiv 1 \pmod{p}$

a $p=7$ $a^6 \equiv 1 \pmod{7}$

$$6614 = 6 \cdot 1102 + 2$$

$$5^{6614} = (5^6)^{1102} \cdot 5^2 \quad 12^{587} = 6 \cdot 97 + 5$$

$$\underbrace{5^6}_{\equiv 5^2 \pmod{7}} \quad 12^{587} = (12^6)^{97} \cdot 12^5 \equiv 12^5 \pmod{7}$$

$$\equiv 5^2 - 12^5 \equiv 1 \pmod{7} \quad -124 = 7 \cdot (-18) + 2 \quad 90 = 7 \cdot 12 + 6$$

$$12 \equiv 5 \pmod{7} \quad 5^2 - 5^5 \Rightarrow 5^2(1 - 5^3) \equiv 5^2(-124) \equiv 5^2(2) \equiv 50 \equiv 1 \pmod{7}$$

b $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$

$p^{q-1} + q^{p-1} \equiv 1 \pmod{p}$ by Fermat's little theorem

$\Rightarrow p^{q-1} + q^{p-1} = kp + 1$ for some $k \in \mathbb{N}$

by symmetry $p^{q-1} + q^{p-1} = nq + 1$ for some $n \in \mathbb{N}$

$nq + 1 = kp + 1 \Rightarrow nq = kp \Rightarrow p|n \text{ \& } q|k$ thus $n = mp$ & $k = jq$

as p & q are distinct primes

$mpq + 1 = p^{q-1} + q^{p-1}$, mod pq this is $\equiv 1$

3 a $\hat{M} \equiv M^k \pmod{n}$

$\equiv 9^7 \pmod{209}$

$9^1 \equiv 9 \pmod{209}$

$9^2 \equiv 81 \pmod{209}$

$9^4 \equiv 6561 \equiv 82 \pmod{209}$

$9^7 \equiv 9 \cdot 81 \cdot 82 \pmod{209}$

$\equiv 102$

$\hat{M} \equiv 4 \pmod{209}$

$(p-1)(q-1) = 180$

b $s \cdot k \equiv 1 \pmod{(p-1)(q-1)}$

$7 \cdot s \equiv 1 \pmod{180}$

$\gcd(7, 180) = 1$

$180 = 25 \cdot 7 + 5$

$7 = 5 \cdot 1 + 2$

$5 = 2 \cdot 2 + 1$

$1 = 5 - 2 \cdot 2$

$1 = 5 - 2(7 - 5) = 3 \cdot 5 - 2 \cdot 7$

$1 = 3(180 - 25 \cdot 7) - 2 \cdot 7$

$3(180) - 77 \cdot 7 \quad s \equiv -77 \pmod{180}$

$\equiv 103$

c $\hat{M}^5 \pmod{209} \quad 209 = 11 \times 19$

$\hat{M} = 4$

$4^{103} \equiv ((4^{10})^{10}) \cdot 4^3 \quad 4^{10} \pmod{11} = 1 \text{ by FLT}$

$\equiv 4^3 \pmod{11}$

$64 \equiv 9 \pmod{11} \quad 64 = 5 \cdot 11 + 9$

$4^{103} = (4^{10})^5 \cdot 4^3 \quad 4^{10} \pmod{19} = 1 \text{ by FLT}$

$\equiv 4^3 \pmod{19} \equiv 9 \pmod{19}$

$13 = 8 + 4 + 1 \quad 5 \cdot 9 \cdot 4 = 180 \quad 180 \equiv 9 \pmod{19}$

$4^{103} = m \cdot 19 + 9 = n \cdot 11 + 9 \Rightarrow 4^{103} \equiv 9 \pmod{209}$ same as 2B $M = 9 \quad \checkmark$

4 a Let e_n = number of n -length sequences ending even

& o_n = number of n -length sequences ending odd

$o_n = 2 \cdot a_{n-1}$, all sequences can be followed by an odd legally, 2 odds

$e_n = 3 \cdot o_{n-1}$, only n -length odd-ending sequences can be followed by an even, 3 evens

$$e_n = 6 \cdot a_{n-2}$$

$$\begin{cases} a_n = o_n + e_n = 2a_{n-1} + 6a_{n-2} \\ a_0 = 1 \\ a_1 = 5 \end{cases}$$

b characteristic polynomial

$$r^2 = 2r + 6 \Rightarrow r^2 - 2r - 6 = 0 \quad \text{roots: } \frac{2 \pm \sqrt{4 - (4)(-6)}}{2} \Rightarrow \frac{2 \pm \sqrt{28}}{2} \Rightarrow 1 \pm \sqrt{7}$$

$$\text{General solution } a_n = A(1+\sqrt{7})^n + B(1-\sqrt{7})^n$$

$$\text{Solving for coefficients } a_0 = A + B = 1$$

$$a_1 = A + \sqrt{7}A + B - \sqrt{7}B = 5$$

$$\begin{pmatrix} (1+\sqrt{7}) & (1-\sqrt{7}) \\ 1 & 1 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$\text{augmented matrix } \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1+\sqrt{7} & 1-\sqrt{7} & 5 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2\sqrt{7} & 4-\sqrt{7} \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -\frac{2}{\sqrt{7}} + \frac{1}{2} \end{array} \right) \rightarrow \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{7}} + \frac{1}{2} \\ -\frac{2}{\sqrt{7}} + \frac{1}{2} \end{pmatrix}$$

$$\text{general solution } a_n = \left(\frac{2}{\sqrt{7}} + \frac{1}{2} \right) (1+\sqrt{7})^n + \left(-\frac{2}{\sqrt{7}} + \frac{1}{2} \right) (1-\sqrt{7})^n$$

check a_0 & a_1

$$a_0 = \frac{2}{\sqrt{7}} + \frac{1}{2} + \frac{-2}{\sqrt{7}} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark$$

$$a_1 = \left(\frac{2}{\sqrt{7}} + \frac{1}{2} \right) (1+\sqrt{7}) + \left(-\frac{2}{\sqrt{7}} + \frac{1}{2} \right) (1-\sqrt{7})$$

$$= \frac{2}{\sqrt{7}} + \frac{1}{2} + 2 + \frac{\sqrt{7}}{\sqrt{7}} + \frac{-2}{\sqrt{7}} + \frac{1}{2} + 2 - \frac{\sqrt{7}}{\sqrt{7}} = \frac{1}{2} + 2 + \frac{1}{2} + 2 = 5 \quad \checkmark$$